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Project Report

Course: EAI-421: Application of AI & ML in Agriculture

Topic : AI based Fertilizer Recommendation system based on soil test values

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ABSTRACT

Agriculture plays a crucial role in sustaining livelihoods and ensuring food security, especially in a country like India. Efficient fertilizer usage is essential for improving crop productivity and maintaining soil fertility. This project presents an AI-based fertilizer recommendation system that predicts optimal nutrient requirements (Nitrogen, Phosphorus, Potassium – NPK) using a Random Forest algorithm.

The system takes inputs such as crop type, soil condition, and irrigation type, and predicts recommended fertilizer doses. It further adjusts recommendations based on real-time soil test values and converts them into practical fertilizer quantities like Urea, DAP, and MOP. This approach ensures precision farming, reduces fertilizer wastage, and enhances sustainable agriculture.

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1. Introduction

Agriculture plays a vital role in India's economy. Proper fertilizer recommendation is essential to improve crop yield and maintain soil health. Traditional methods of fertilizer application are often based on assumptions, which may lead to overuse or underuse of nutrients.

This project presents a machine learning-based system that recommends optimal fertilizer quantities (N, P, K) based on crop type, soil type, and irrigation conditions (irrigated/rainfed).

The system uses a dataset of 50 major crops grown in India with their Recommended Dose of Fertilizers (RDF). A Random Forest Regressor model is trained to predict nutrient requirements, which are further adjusted based on soil nutrient values. The system also converts nutrient recommendations into real fertilizers such as Urea, DAP, and MOP.

This intelligent system helps farmers make better decisions, improving productivity and reducing environmental impact.

2. Problem Statement

In many regions, farmers rely on traditional knowledge or guesswork for fertilizer application. This leads to:

- Imbalanced nutrient usage
- Soil degradation
- Increased farming costs
- Reduced crop yield

There is a need for a smart system that provides precise fertilizer recommendations based on scientific data and soil conditions.

3.Motivation

The motivation behind this project is to address real-world challenges faced by farmers in fertilizer management. Many farmers lack access to precise recommendations, which leads to inefficient use of resources.

This system is developed with the following motivations:

- Helps farmers reduce cultivation costs and increase overall profit
- Promotes sustainable agriculture by avoiding excessive fertilizer use
- Prevents soil degradation caused by overuse of chemicals
- Encourages adoption of smart farming techniques using Artificial Intelligence
- Bridges the gap between traditional farming and modern technology

Relevance of the Project:

- Agricultural research and development
- Agri-tech startups focusing on precision farming
- Government farming initiatives and subsidy programs

4. Objectives

- To develop an AI-based fertilizer recommendation system
- To predict accurate NPK values using machine learning
- To assist farmers in efficient fertilizer usage
- To reduce environmental damage caused by excessive fertilizers
- To convert nutrient values into real-world fertilizers
- To support sustainable agriculture practices

5. Methodology

5.1 Data Collection

The dataset contains:

- Crop name
- Soil/condition type (irrigated/rainfed)
- Recommended N, P, K values

It includes data for around 50 crops commonly grown in India.

5.2 Data Preprocessing

- Convert categorical variables (crop, condition) into numeric format
- Use Label Encoding for transformation
- Prepare dataset for machine learning model

5.3 Model Training

- Algorithm used: Random Forest Regressor
- Input features:
  - Crop
  - Condition
- Output:
  - N, P, K values

The model learns patterns from the dataset to predict fertilizer requirements.

5.4 Prediction Process

- User selects crop and condition
- Inputs soil nutrient values
- Model predicts base NPK values

5.5 Adjustment Based on Soil

The predicted values are adjusted using soil test inputs:

- If soil has less nutrients → recommendation increases
- If soil has more nutrients → recommendation decreases

This ensures accurate fertilizer usage.

5.6 Fertilizer Conversion

The system converts nutrients into real fertilizers:

- **Urea** → Nitrogen source
- **DAP** → Phosphorus source
- **MOP** → Potassium source

This makes the output practical for farmers.

6. Technologies Used

- Python (Programming Language)
- Pandas (Data handling)
- Scikit-learn (Machine Learning)
- Random Forest Algorithm

7. System Architecture

```
User Input → Data Preprocessing → ML Model (Random Forest)
→ NPK Prediction → Adjustment → Fertilizer Conversion → Output
```

8. Algorithm Used: Random Forest

Random Forest is a supervised machine learning algorithm that:

- Uses multiple decision trees
- Combines outputs for better accuracy
- Handles nonlinear data effectively
- Reduces overfitting

It is highly suitable for agricultural prediction systems.

9.CODE

```
import pandas as pd

from sklearn.ensemble import RandomForestRegressor

from sklearn.preprocessing import LabelEncoder


# ----------------------------

# LOAD DATASET

# ----------------------------

df = pd.read_csv("crop_rdf_data.csv")


# ----------------------------

# ENCODE COLUMNS

# ----------------------------

le_crop = LabelEncoder()

le_condition = LabelEncoder()


df['crop_encoded'] = le_crop.fit_transform(df['crop'])

df['condition_encoded'] =
le_condition.fit_transform(df['condition'])


# ----------------------------

# TRAIN MODEL

# ----------------------------
```

```
X = df[['crop_encoded', 'condition_encoded']]
y = df[['N', 'P', 'K']]

model = RandomForestRegressor()
model.fit(X, y)

# ----------------------------
# DISPLAY CROPS
# ----------------------------
crop_list = list(le_crop.classes_)

print("\nAvailable Crops:")
for i, c in enumerate(crop_list, start=1):
    print(f"{i}. {c.capitalize()}")

# ----------------------------
# USER INPUT (CROP) WITH LOOP
# ----------------------------
while True:
    try:
        crop_index = int(input("\nEnter crop number: "))
        if 1 <= crop_index <= len(crop_list):
            crop = crop_list[crop_index - 1]
```

```
            break
        else:
            print("Error: Index out of range. Please enter a valid crop number.")
    except:
        print("Error: Please enter a numeric value.")

# ----------------------------
# DISPLAY CONDITIONS
# ----------------------------
condition_list = list(le_condition.classes_)

print("\nConditions:")
for i, cond in enumerate(condition_list, start=1):
    print(f"{i}. {cond.capitalize()}")

# ----------------------------
# USER INPUT (CONDITION) WITH LOOP
# ----------------------------
while True:
    try:
        cond_index = int(input("\nEnter condition number: "))
        if 1 <= cond_index <= len(condition_list):
```

```
        condition = condition_list[cond_index - 1]
        break
    else:
        print("Error: Index out of range. Please enter a valid condition number.")
  except:
    print("Error: Please enter a numeric value.")

# ----------------------------
# SOIL INPUT WITH LOOP
# ----------------------------
while True:
  try:
    N = float(input("Enter soil N: "))
    P = float(input("Enter soil P: "))
    K = float(input("Enter soil K: "))
    break
  except:
    print("Error: Please enter valid numeric values for N, P, K.")

# ----------------------------
# PREDICTION
# ----------------------------
```

```
crop_code = le_crop.transform([crop])[0]
cond_code = le_condition.transform([condition])[0]

base = model.predict([[crop_code, cond_code]])
rec_N, rec_P, rec_K = base[0]

# ----------------------------
# ADJUSTMENT
# ----------------------------
final_N = rec_N + (300 - N) * 0.2
final_P = rec_P + (20 - P) * 0.3
final_K = rec_K + (200 - K) * 0.1

# ----------------------------
# FERTILIZER CONVERSION
# ----------------------------
dap = final_P / 0.46
N_from_dap = dap * 0.18

remaining_N = final_N - N_from_dap
urea = remaining_N / 0.46

mop = final_K / 0.60
```

```
# ----------------------------
# OUTPUT
# ----------------------------
print("\n===== FINAL RECOMMENDATION =====")
print(f"Crop: {crop.capitalize()}")
print(f"Condition: {condition.capitalize()}")

print("\nNPK (kg/ha):")
print("N:", round(final_N, 2))
print("P:", round(final_P, 2))
print("K:", round(final_K, 2))

print("\nFertilizers (kg/ha):")
print("Urea:", round(max(0, urea), 2))
print("DAP:", round(max(0, dap), 2))
print("MOP:", round(max(0, mop), 2))
```

10.Advantages

- Provides accurate fertilizer recommendations
- Reduces fertilizer wastage
- Improves crop productivity
- Easy to use for farmers
- Supports precision agriculture

11.Limitations

- Depends on dataset quality and size
- Requires proper soil testing
- Limited to trained crops only
- May not consider weather conditions

12. Future Enhancements

- Mobile application for farmers
- Integration with IoT soil sensors
- Weather-based recommendations
- Government soil database integration
- Multi-language support

13. Practical Applications

This system can be applied in multiple real-world scenarios:

- **Smart Farming Systems:**
  Helps farmers make data-driven decisions for fertilizer usage
- **Government Agriculture Schemes:**
  Can be integrated into soil health card programs and advisory systems
- **Agri-tech Startups:**
  Useful for developing intelligent farming applications and advisory platforms
- **Educational Tools:**
  Can be used for teaching machine learning applications in agriculture

14. Learning Outcomes

Through this project, the following knowledge and skills are gained:

- Understanding of machine learning concepts and algorithms
- Practical implementation of the Random Forest model
- Data preprocessing and feature encoding techniques
- Handling real-world agricultural datasets
- Developing solutions for real-world problems using AI
- Knowledge of fertilizer composition and agricultural practices

15. Future Improvements

The current system can be further enhanced in the following ways:

- **Add Weather and Climate Data:**
  Include rainfall, temperature, and humidity for better predictions
- **Use Larger Real-World Datasets:**
  Improve accuracy by training on more diverse and extensive datasets
- **Develop Mobile/Web Application:**
  Create a user-friendly interface for farmers
- **Integrate IoT-Based Soil Sensors:**
  Automatically collect real-time soil data
- **Use Advanced Models (Deep Learning):**
  Improve prediction accuracy with neural networks and advanced AI techniques

16. Conclusion

This project demonstrates how artificial intelligence can be applied in agriculture to provide precise fertilizer recommendations. By using machine learning and soil data, farmers can improve crop yield, reduce costs, and maintain soil health. The system supports sustainable and modern farming practices.